Prediction of Cardiovascular Outcomes Using ECG from Overnight Polysomnography

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ABSTRACT

Study Objectives: Polysomnography (PSG) is the gold standard for diagnosing sleep disorders, which are increasingly recognized as cardiovascular risk factors. Single-lead electrocardiograms (ECGs) are routinely recorded during PSG but rarely analyzed, presenting an opportunity to leverage overnight ECGs for long-term cardiovascular risk prediction.

Methods: We applied a deep residual neural network with attention to single-lead ECGs from PSG, combined with expert-annotated sleep stage data, to predict 10-year risk of atrial fibrillation (AF), stroke, myocardial infarction (MI), heart failure (HF), and death. The network, originally trained for arrhythmia detection, was fine-tuned on 15,809 Massachusetts General Hospital patients. External validation used cohorts from Emory University Hospital (EUH) (n=9,810) and Beth Israel Deaconess Medical Center (BIDMC) (n=12,576), with outcomes derived from electronic health records (ICD-9/10 codes).

Results: Cox proportional hazards models showed that neural-network output (NN-output) was strongly associated with long-term cardiovascular risk ($p < 0.0001$), even after adjusting for traditional risk factors (age, sex, body mass index, smoking, hypertension, diabetes) and sleep characteristics (apnea-hypopnea-index, arousal-index, periodic limb movement index, time spend in N1-N3, REM and sleep efficiency). A one-standard deviation increase in NN-output was associated with elevated hazard ratios: AF 2.03 (EUH) and 2.72 (BIDMC); stroke 1.19 and 1.40; MI 1.38 and 1.20; HF 1.69 and 2.33; all-cause mortality 1.82 and 1.65.

Conclusion: Deep learning applied to nocturnal ECGs with sleep stage information captures subtle cardiac abnormalities, offering a scalable, non-invasive tool for cardiovascular risk stratification. Integrating such prediction into PSG could enhance early detection and support personalized, cost-effective clinical decision-making.

Keywords

electrocardiogram (ECG), polysomnography (PSG), risk prediction, cardiovascular disease, heart failure, atrial fibrillation, myocardial infarction, stroke

Statement of significance

We present a neural network that integrates overnight ECG signals with expert-annotated sleep stages to predict incident cardiovascular outcomes in patients undergoing routine polysomnography. Using large-scale sleep study data, the model predicts atrial fibrillation, stroke, myocardial infarction, heart failure, and death up to ten years after the sleep study. While traditional deep-learning-based risk prediction models focus on standard 12-lead ECGs, the ECG channel in polysomnography is typically underutilized. Given the close relationship between sleep disorders and cardiovascular disease, this work addresses a largely unexplored opportunity for risk prediction in sleep study populations. Consistent performance across multiple sleep-study cohorts demonstrates generalizability, supporting deep-learning applied to overnight ECG combined with sleep-stage information as a non-invasive tool for long-term cardiovascular outcome prediction.

Introduction

Polysomnography (PSG) is the standard diagnostic method for evaluating many sleep related disorders. Sleep disorders, including sleep-disordered breathing, or insomnia, have been linked with the incidence of heart disease.[1–3] Research has shown that individuals with obstructive sleep apnea (OSA) have a higher prevalence of atrial fibrillation (AF), a condition that increases the risk of stroke fivefold and the risk of heart failure (HF) threefold compared to the general population[4]. The atrial arrhythmogenic mechanism in patients with OSA is largely attributed to autonomic nervous system imbalances, chronic intermittent hypoxia, inflammation, and swings in thoracic pressure during different phases of the acute apnoeic episodes[3]. During the apnoeic phase itself, there is substantial bradycardia, signaling parasympathetic dominance[5]. Following an apnea event, a pronounced rebound tachycardia occurs, likely due to parasympathetic withdrawal, which uncovers an increase in cardiac sympathetic activity[2]. These disruptions often lead to elevated heart rates and blood pressure in individuals with OSA, placing increased stress on the cardiovascular system. This prolonged strain promotes the development of coronary artery disease and heightens the risk of myocardial infarction (MI), stroke, and HF.

Insomnia is also strongly associated with cardiovascular disease[6–8]. Insomnia can be a symptom of a wide range of psychiatric and medical conditions, or arise due to environmental factors including stress. When insomnia becomes chronic and severe, it can itself act as a stressor, impacting the patient's daily life. In individuals with chronic insomnia, dysregulation of the hypothalamic-pituitary-adrenal (HPA) axis leads to elevated levels of adrenocorticotropic hormone (ACTH) and cortisol[6]. Elevated cortisol levels are linked to an increased risk of MI[9]. Chronic stress and prolonged cortisol elevation accelerate atherosclerosis, further increasing the risk of MI.

The overlap between PSG data and cardiovascular risk present a unique opportunity to predict cardiovascular outcomes using PSG. Given that many patients undergoing PSG are at higher risk for cardiovascular events, the data collected from their sleep studies are particularly valuable. During PSG, multiple physiological signals are recorded, including electrocardiography (ECG), which provides information about cardiac function, autonomic activity, and arrhythmia burden throughout sleep. Traditional cardiovascular risk models primarily rely on awake or ambulatory ECG recordings, but analyzing nocturnal ECG from PSG may reveal additional risk markers influenced by sleep-stage transitions, oxygen desaturations, or autonomic fluctuations. In this study, we develop a deep learning algorithm to predict multiple cardiovascular outcomes—atrial fibrillation (AF), stroke, myocardial infarction (MI), heart failure (HF), and all-cause mortality, using the ECG signal combined with sleep stages from PSG. By leveraging sleep-related cardiac dynamics, we aimed to determine whether information extracted from nocturnal ECG using deep learning offers independent prognostic value beyond conventional clinical risk factors, potentially improving early risk stratification and informing personalized treatment strategies.

Dataset

In this study, we analyzed three independent datasets collected from sleep laboratories at Massachusetts General Hospital (MGH), Emory University Hospital (EUH), and Beth Israel Deaconess Medical Center (BIDMC) obtained as part of the Human

Sleep Project[10]. This retrospective analysis was conducted under IRB protocol #2013P001024 at MGH, which also granted a waiver of consent. The MGH PSGs, were used for model development and internal validation. The dataset contains 26,369 recordings, of which 23,259 recordings from 17,328 subjects included an ECG channel and were included in our analysis. The mean ECG duration was 7.62 ± 1.34 hours.

The EUH dataset served as an independent test set and consists of 14,830 PSGs from 12,067 patients, each containing a single ECG channel. The mean duration of the ECG recordings was 7.20 ± 1.25 hours.

The BIDMC dataset was also used as an external test set and includes 12,976 PSGs from 12,576 patients, each with a single ECG channel. The mean ECG duration was 7.70 ± 1.22 hours.

All recordings followed American Academy of Sleep Medicine (AASM) standards[11]. For outcome prediction, we used a single ECG lead placed below the right clavicle near the sternum and over the left lateral chest wall. Signals were stored in microvolts and volts with sampling rates of 200, 250, 500, 512 or 1,000 Hz and were resampled to 500 Hz for analysis.

Cardiovascular Outcomes

This study focuses on the prediction of cardiovascular outcomes. We extracted five outcomes: atrial fibrillation (AF), stroke, myocardial infarction (MI), heart failure (HF), and all-cause death. All outcomes except death were collected based on ICD-9/ICD-10 codes from electronic health records (EHR); death and the date of death was collected from the EHR as well. The list of codes for each diagnosis is specified in Table 1.

In the next sections, when performing survival analysis, we focus on incident (new) cases of each cardiovascular outcome by excluding individuals with a prior diagnosis of the respective condition. For example, if predicting AF, we ensure that the AF survival analysis model does not include patients diagnosed with AF before the PSG date. Specifically for atrial fibrillation, we also excluded subjects diagnosed with AF within one month after the PSG date, because patients with AF first detected or suspected on PSG would likely receive an AF ICD code in the days to weeks following PSG, if this was clinically identified.

For patients who eventually tested positive for an outcome, we extracted the time-to-event (in years) based on the time from the PSG date to the first mention of the diagnosis (ICD code) in the EHR. For patients without a positive outcome for a specific disease, ending time for the analysis (censoring date) was extracted based on the last recorded date for the patient in the EHR. The total number of subjects and the number of subjects with future outcomes, along with mean time-to-event for each outcome, are presented in Table 2.

Outcome	ICD-9	ICD-10
Atrial Fibrillation	427.3	I48
Stroke	430, 431, 433, 434, 436	I60, I61, I63, I64
Myocardial Infarction	410, 411, 412	I21, I22, I23
Heart Failure	428	I50

Table 1. Outcomes and corresponding ICD-9 and ICD-10 codes used for CVD outcome extraction from the electronic health records.

Atrial Fibrillation Label Correction

Because some patients may have undiagnosed AF within the hospital system or paroxysmal AF not yet diagnosed, and because our goal is to predict future AF rather than detect existing but unrecognized cases, we implemented additional preventative measures. Manual review of the dataset—over 300,000 hours of PSG recordings—was not feasible. Instead, we applied two automated AF classification algorithms: one from the PhysioNet Challenge 2017[12] and another from the PhysioNet Challenge 2021[13]. Both algorithms analyzed ECG data in 30-second windows, classifying each segment as AF or non-AF.

To manage the data scale, we restricted this analysis to patients with future AF incidence but no prior AF history. For PSGs flagged as AF by either algorithm, we selected the five 30-second ECG segments with the highest AF probability and forwarded these to a cardiologist for review. If AF was confirmed by cardiologist in any segment, the patient was reclassified as having current AF and excluded from the future AF prediction analyses. This review procedure was applied to both the MGH validation set and the test set. For the MGH validation set, 67 ECGs were flagged by the algorithms, with 29 confirmed as AF by the cardiologist. For the test set, 119 ECGs were flagged, with 42 confirmed as AF. We believe that this approach, reviewing all recordings flagged by at least one algorithm, captures a meaningful subset of AF-positive cases.

Covariates

From the demographic data, we extracted each patient's age at the time of the PSG and sex. Body mass index (BMI) was obtained when available on the day of the PSG; if not available, we used measurements recorded up to one year prior, given BMI's established association with cardiovascular disease. Additional demographic details are provided in Table 2.

From the sleep and sleep-breathing metrics, we extracted time spent in each sleep stage (N1, N2, N3, and REM), sleep efficiency, the apnea–hypopnea index (AHI), the periodic limb movement index (PLMI), and the arousal index (ArI), all derived from overnight PSGs based on expert annotations. Mean values and standard deviations for these metrics across the cohorts are summarized in Table 3.

Given their relevance to cardiovascular outcomes, we extracted comorbidity information from EHR-derived ICD-9/10 codes. Specifically, we identified diabetes using codes E08.x, E09.x, E10.x, E11.x, E13.x, and 250.x; hypertension using codes I10.x, I11.x, I12.x, I13.x, I15.x, 401.x, 402.x, 403.x, 404.x, and 405.x; and smoking history using codes 305.1x, V15.82, F17.x, Z72.0, Z87.891, and T65.2x. The number and percentage of subjects with these conditions are reported in Table 4. All of these covariates (demographics, sleep metrics and comorbidity information) were included in the survival analysis models.

Patients were included in the final study cohort if they met the following criteria: (1) ECG available during the PSG, (2) a documented event or last follow-up date, (3) available expert sleep scoring, and (4) available BMI. After applying these criteria, the final datasets consisted of 12,381 patients in the MGH training set, 3,428 in the MGH validation set, 9,810 in the EUH test set, and 12,576 in the BIDMC test set.

Outcome	No of subjects (%)[a]	Time to event[b,c]	Sex (% female)	Age [b,d]	BMI[b]
Training - MGH					
Overall	12,381	-	43.4	52.9 (16.6)	30.55 (7.67)
Atrial Fibrillation	1,208 (9.76)	4.6 (3.2)	32.1	65.0 (11.8)	32.17 (7.53)
Stroke	987 (7.97)	5.1 (3.4)	41.7	62.6 (13.7)	30.70 (6.84)
Myocardial Infarction	341 (2.75)	5.5 (3.4)	29.8	62.9 (11.4)	32.60 (7.18)
Heart Failure	1,099 (8.88)	5.0 (3.4)	38.7	65.3 (12.6)	33.48 (8.14)
Death	827 (6.68)	5.3 (3.3)	35.2	67.5 (13.5)	31.92 (7.61)
Validation - MGH					
Overall	3,428	-	42.5	53.2 (16.7)	30.49 (7.58)
Atrial Fibrillation	368 (11.25)	4.7 (3.2)	33.7	65.3 (12.6)	32.08 (7.38)
Stroke	312 (9.44)	4.9 (3.3)	41.9	62.6 (13.0)	31.21 (7.33)
Myocardial Infarction	82 (2.44)	5.1 (3.2)	32.1	64.5 (11.0)	31.88 (5.86)
Heart Failure	313 (9.51)	4.9 (3.4)	41.8	65.0 (13.6)	34.32 (8.67)
Death	237 (7.02)	5.2 (3.3)	30.3	67.9 (14.1)	31.45 (7.62)
Test - EUH					
Overall	9,810	-	51.2	54.7 (15.5)	31.32 (14.51)
Atrial Fibrillation	605 (7.12)	2.8 (2.4)	46.5	62.4 (13.2)	31.74 (16.41)
Stroke	781 (9.39)	2.8 (2.4)	48.6	62.1 (12.8)	31.78 (19.05)
Myocardial Infarction	464 (5.32)	2.7 (2.3)	49.3	60.5 (13.2)	31.83 (12.74)
Heart Failure	832 (10.35)	2.7 (2.3)	52.5	61.5 (13.9)	32.17 (13.30)
Death	498 (4.88)	3.5 (2.4)	43.3	64.8 (13.8)	31.29 (14.51)
Test - BIDMC					
Overall	12,576	-	45.3	56.8 (16.9)	33.52 (8.95)
Atrial Fibrillation	409 (4.7)	1.9 (1.7)	34.2	67,3 (12.3)	31.76 (7.10)
Stroke	180 (1.9)	1.9 (1.6)	35.6	66.7 (12.4)	33.04 (7.51)
Myocardial Infarction	130 (1.4)	2.2 (1.7)	36.9	67.7 (12.5)	32.93 (8.67)
Heart Failure	520 (6.1)	1.9 (1.7)	44.0	67.0 (12.2)	32.72 (8.79)
Death	378 (3.00)	2.1 (1.3)	38.9	70.1 (12.3)	32.71 (8.42)

Table 2. Data characteristics.

[a] Number of participants for each outcome category
[b] Mean with standard deviation in parentheses.
[c] In years.
[d] At the time of the PSG.

Methods

Outcome Prediction using a Deep Neural Network

The aim of this study is to predict cardiological outcomes using a deep learning approach. We leverage the ECG channel recorded during PSG together with expert-annotated sleep stages. First, a pretrained encoder is used to extract feature representations from the ECG signal, which are then combined with sleep-stage information for cardiovascular outcome prediction. Figure 1 provides a schematic overview of the model architecture: panel A shows the pretrained CNN encoder, and panel B illustrates the newly added layers for outcome prediction.

The pretrained encoder is based on the winning solution from the George B. Moody PhysioNet Challenge 2021[14], a residual neural network with an attention mechanism[13] designed for arrhythmia classification. This model was pretrained on 86,000 ECG recordings for the classification of 26 arrhythmias. Although designed for 12-lead clinical ECGs, its architecture supports

	MGH - training	MGH - validation	EUH	BIDMC
Time in N1 (min)	51.26 (39.61)	51.34 (39.84)	24.80 (18.75)	43.96 (28.31)
Time in N2 (min)	190.99 (72.54)	191.44 (74.12)	197.80 (60.29)	187.90 (69.09)
Time in N3 (min)	54.25 (41.36)	52.95 (40.80)	39.13 (37.51)	30.58 (32.32)
Time in REM (min)	56.03 (35.42)	55.51 (35.70)	52.07 (33.01)	53.23 (36.67)
Sleep efficiency (%)	76.70 (19.17)	76.58 (19.14)	74 (16)	68 (17)
AHI	6.74 (27.28)	7.29 (32.51)	6.35 (8.00)	22.98 (20.24)
PLM index	18.50 (91.91)	17.74 (39.57)	7.36 (19.77)	13.65 (23.66)
ArI	13.38 (16.28)	14.51 (64.72)	26.75 (18.51)	34.48 (20.68)

Table 3. Sleep architecture and sleep-breathing metrics across cohorts.

	MGH - training	MGH - validation	EUH	BIDMC
Diabetes (%)	2,157 (12.2)	559 (12.6)	2,937 (29.5)	2,375 (18.3)
Hypertension (%)	5,548 (31.3)	1,330 (30.0)	6,490 (65.3)	5,405 (41.7)
Smoking (%)	1,637 (9.2)	422 (9.5)	2,429 (24.4)	2,410 (18.6)

Table 4. Prevalence of diabetes, hypertension, and smoking in each cohort

between 1 and 12 input channels by zero-padding any missing leads.

The neural network takes a 15-second ECG segment as input. The preprocessing steps included resampling all signals to 500 Hz, applying a bandpass filter from 0.05 to 47 Hz, and performing z-score normalization. The neural network input consists of a 12 × 8,192 array, representing a 15-second ECG segment sampled at 500 Hz (zero-padded to 8,192 samples). Because PSG recordings contain only a single ECG lead, this signal was mapped to the lead II position, while all remaining leads were zero-padded to conform to the required 12-lead input format and maintain a consistent input shape. We used the 2021 PhysioNet Challenge residual network as a pretrained encoder (Figure 1A), extracting 1 x 256-dimensional embeddings from each 15-second segment. At this stage, no additional training was performed on the sleep-study ECGs; the model was used solely as a feature extractor.

To adapt the pretrained encoder for outcome prediction, we introduced a long short-term memory (LSTM) based classifier. For each 15-second PSG segment, we combined the extracted 1×256 ECG embedding with a one-hot encoded vector of size 1×5 representing the expert annotated sleep stage (Wake, REM, N1, N2, N3) from PSG annotations. This resulted in an input tensor of shape (1 × number of 15-second segments × 261), which was passed through the LSTM module, followed by an attention mechanism and a classification head composed of fully connected layers, ReLU activation, DropOut, and a final sigmoid layer. The model outputs the probability of five non-mutually exclusive outcomes: future AF, stroke, MI, HF, and death. The LSTM classifier (Figure 1B) was trained directly on the sleep study ECG recordings.

For model development, the MGH dataset was divided into training and validation subsets. The training set included 18,592 ECG recordings from 12,381 subjects, and the validation set included 4,667 recordings from 3,428 distinct subjects, with no subject overlap between splits. The EUH and BIDMC cohorts served as the external test set.

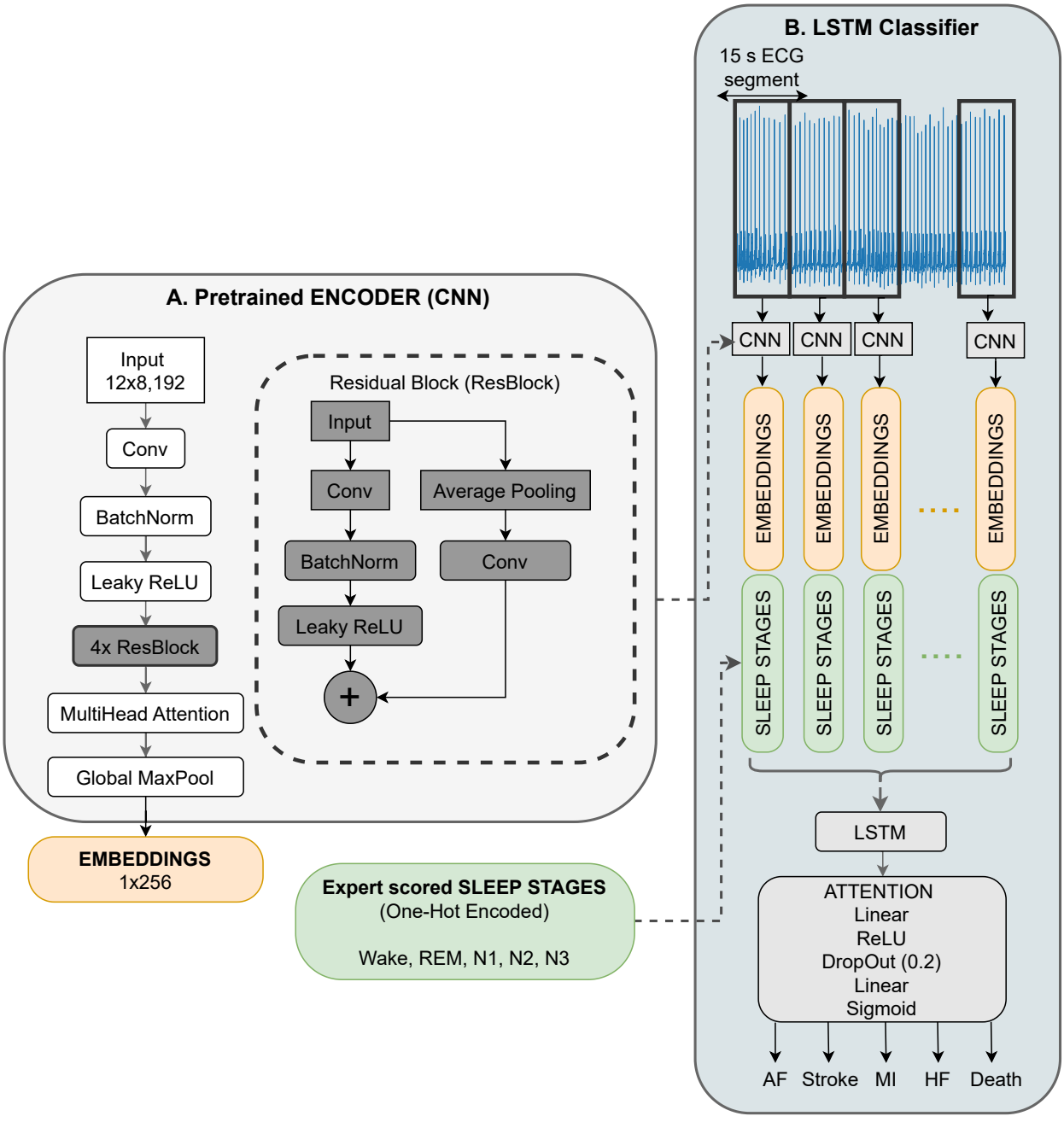


Figure 1. Neural network architecture for cardiology outcome prediction. Part A. illustrates the pretrained embeddings extractor from the PhysioNet Challenge 2021 winning solution, while the B section represent newly added layers for prediction of 5 non-mutually exclusive CVD outcomes combining ECG embeddings with one-hot encoded sleep stages.

Training and Inference

During training, full-night PSG recordings were segmented into consecutive 15-second ECG windows. For each window, we generated a 1×256 ECG embedding and appended the corresponding one-hot encoded sleep stage vector. The sequence of

these feature vectors across the entire night was fed into the LSTM classifier. We trained the model using the Adam optimizer with a learning rate of 0.001 and a learning rate scheduler with a step size of 5. Training was performed for 20 epochs with early stopping based on the area under the receiver operating characteristic curve (AUROC) and area under the precision-recall curve (AUPRC) on the validation set to prevent overfitting. The loss function was binary cross-entropy, weighted by (1 - class prevalence) to mitigate class imbalance across the five outcomes. During inference, the pipeline mirrored the training procedure.

Cox proportional hazard model

To evaluate how well the neural network output derived from ECG and sleep-stage information predicts cardiovascular outcomes, we performed a survival analysis. In survival analysis we focus on incident (new) cases of each cardiovascular outcome by excluding individuals with a prior diagnosis of the respective condition. Because Cox regression assumes independence of observations, we further included only the most recent PSG recording for subjects with multiple studies.

To visualize the relationship between NN-output and clinical outcomes, we performed a time-to-event analysis. Patients were stratified into three risk categories based on the 20th and 80th percentiles of the NN-output distribution from the MGH validation dataset, applied to the EUH and BIDMC test sets: low risk (below the 20th percentile), medium risk (20th–80th percentile), and high risk (above the 80th percentile). We computed non-parametric cumulative incidence curves (CICs) for each NN-output risk group, derived directly from the observed data. These curves reflect the unadjusted relationship between NN-output and outcomes. Cumulative incidence functions were estimated using the Aalen–Johansen estimator[15], treating death as a competing risk for all outcomes except death itself, which was estimated as one minus the Kaplan–Meier survival function.

For the Cox regression analysis, we trained the model on the MGH validation set and evaluated it on independent test sets from EUH and BIDMC. The maximum follow-up duration was 10 years for the EUH cohort and 7 years for the BIDMC cohort. For all outcomes except death, we used multi-state Cox regression[16,17] to account for death as a competing risk, since death precludes the occurrence of non-fatal outcomes such as AF, stroke, MI, or HF. This approach allows for proper modeling of the transition from the initial state (no event) to either the event of interest or the competing event (death), thus providing unbiased estimates of the covariate effects on each outcome. For the death outcome itself, we used cause-specific Cox regression, as there is no competing risk when death is the terminal event. A separate Cox regression model is fitted for each outcome.

The Cox proportional hazards regression model can be expressed as: $h(t \mid X) = h_0(t) \exp(\beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_p X_p)$ where $h(t \mid X)$ is the hazard at time t for an individual, $h_0(t)$ is the baseline hazard, and $\beta_1, \beta_2, ..., \beta_p$ are the coefficients. The computed risk is given by the linear predictor $\eta = \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p$, which is used to estimate the likelihood of event occuring.

The input variables for the Cox regression include the output of the deep learning model, referred to as the neural network output (NN-output), along with the following covariates: age, sex, BMI, AHI, ArI, PLMI, time spend in N1, N2, N3, REM, sleep efficiency, history of diabetes, smoking and hypertension. The NN-output is first transformed using a logit function and then standardized with a z-score normalization. Sex is kept as the binary variable, with male equal to 1 and the rest of the variables are also standardized with a z-score normalization. For consistent preprocessing across datasets, means and standard deviations computed from the MGH set are used for standardizing the test set.

To evaluate the prognostic value of the NN-output, we performed a multivariate Cox proportional hazards regression, adjusting for relevant covariates. From this model, we extracted the adjusted hazard ratio (HR) associated with the continuous NN-output. Since the NN-output is standardized using z-scores, the HR corresponds to the effect of a one–standard deviation increase in NN-output on the hazard of the outcome.

Calibration between predicted risk from the Cox regression model based on NN-output and the observed risk was evaluated using calibration curves over a 6-year follow-up period. A 6-year horizon was chosen to allow a fair comparison between the EUH and BIDMC cohorts, as the BIDMC has a shorter maximum follow-up of up to 7 years. These curves were generated by fitting a spline to the calibration data with the UnivariateSpline function from the SciPy Python library (version 1.5.4) using a smoothing factor of 1. Calibration performance was assessed using the estimated calibration index (ECI), defined as the root mean squared difference between the mean predicted probabilities and the spline-fitted calibration curve, with lower values indicating better calibration.

Decision curve analysis (DCA) was performed by plotting the net benefit (NB) across a range of decision thresholds. The NB at a given threshold t was computed as: $NB = \frac{TP}{N} - \frac{FP}{N}\frac{t}{1-t}$, where TP and FP denote the number of true and false positives, respectively, and N is the total number of patients.

In a real-world deployment scenario, a predictive model would be used to guide downstream cardiovascular screening among individuals undergoing sleep studies. Therefore, model performance was compared with two default strategies: *Screen None* and *Screen All*. The *Screen None* strategy yields a net benefit of 0 because both TP and FP are zero. The *Screen All* strategy corresponds to screening the entire population, with the costs associated with over-screening (e.g., false positives, resource use). For this strategy, TP/N is replaced by the event prevalence and FP/N by $1-$ prevalence. For example, with a 5% AF rate, *Screen All* implies that 5% of individuals are correctly classified and 95% are incorrectly classified at any given threshold.

For each cardiovascular outcome, net benefit was computed for two models: a Cox regression model using covariates only and a Cox regression model incorporating both covariates and the NN-output. Sensitivity and specificity were then evaluated at the threshold that maximized net benefit when comparing the covariates-only and covariates + NN-output models; this threshold was considered the optimal decision threshold. Sensitivity and specificity at a 6-year follow-up at a given threshold were reported only for outcomes in which the inclusion of NN-output improved net benefit.

Finally, we investigated which 15-second ECG segments contributed most strongly to the model's predictions by analyzing the attention weights. Because the attention mechanism follows the LSTM layer, each 15-second embedding vector is assigned an attention weight reflecting its relative importance. For each subject, we computed the mean attention weight within each sleep stage (wake, REM, and N1–N3). We then compared these distributions across outcome groups to determine the sleep stage during which the model placed the greatest emphasis on the ECG segments.

We also assessed the relationship between the NN-output and the clinical covariates included in the Cox regression by computing Spearman correlation coefficients.

Sensitivity analysis

Because our analysis relies on predicting incident CVD events using ICD codes, it is important to account for potential noise and misclassification in the EHR. To reduce the impact of mislabeling, we implemented several preventive measures.

First, for all CVD outcomes except death (which is not defined by ICD codes), we required that incident cases have at least two ICD codes corresponding to the condition of interest. Individuals with only a single ICD code for AF, stroke, MI, or HF were excluded to minimize the likelihood of false-positive outcome assignments.

Second, we performed an additional sensitivity analysis for AF prediction. Patients with paroxysmal AF may remain undiagnosed for extended periods if they are asymptomatic, raising the possibility that some individuals labeled as AF-free at baseline may actually suffer from AF at the time of the sleep study or develop AF shortly after. Our primary analysis removed subjects diagnosed with AF within one month after the PSG and applied the correction pipeline. To further mitigate potential misclassification, we repeated the analysis after excluding all patients diagnosed with AF within 6 months and again within 12 months following the PSG.

Finally, we conducted a sub-analysis restricted to patients with no prior history of AF, stroke, MI, or HF. While earlier analyses for specific outcomes (e.g., MI incidence) excluded patients with a history of the outcome of interest (e.g., prior MI for MI prediction), they allowed inclusion of patients with other cardiovascular conditions. In contrast, this analysis excluded all individuals with any history of the four cardiovascular diseases (AF, stroke, MI, or HF). The rationale behind this approach is to examine the prediction of cardiovascular outcomes specifically in a cohort of individuals without any previous history of cardiovascular disease (included in the list). By doing so, we aim to assess the ability of the model to predict cardiovascular events in a generally healthier population, distinct from patients who may already have existing cardiovascular conditions or are receiving treatment for them. This allows us to focus on predicting outcomes in a group of individuals who have not yet experienced significant cardiovascular events, thereby investigating the predictive power of the model in a population at lower baseline risk.

Results

Unadjusted cumulative incidence curves (1 – Kaplan-Meier) in Figure 2 consistently demonstrated higher event rates among subjects with medium- and high-risk NN-output compared with those in the low-risk category (log-rank $p < 0.0001$) across all cardiovascular outcomes.

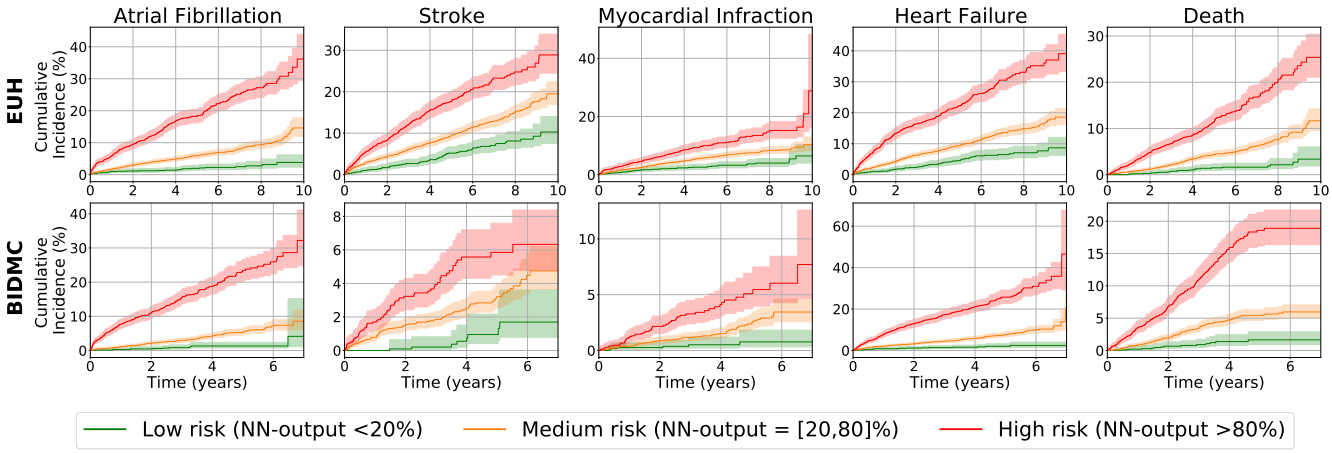


Figure 2. Cumulative incidence curves for five cardiovascular outcomes stratified by NN-output risk category at the EUH site (first row) and BIDMC site (second row). Differences between low- and medium-risk, as well as low- and high-risk groups, were statistically significant for all outcomes (log-rank test p <0.0001).

The results of the Cox regression analysis on the test set, including covariate-adjusted hazard ratios (HRs) with 95% confidence intervals (CIs) for the NN-output, are presented in Table 5 for both EUH and BIDMC site. The hazard ratios reflect the effect of a one–standard deviation increase in the NN-output. Across all outcomes, higher NN-output values are associated with an increased risk of cardiovascular events, indicating that the neural network predictions are consistently aligned with elevated event risk. For the MI outcome, the hazard ratio in the BIDMC cohort was not statistically significant, suggesting weaker predictive performance of the deep learning model; however, this may also be attributable to the very low MI prevalence (1.39%) in the BIDMC cohort.

Cohort	Outcome	Positive outcome (%)	Hazard ratios (95% CI)
EUH	Atrial Fibrillation	605 (7.12)	2.03 (1.82–2.27) ****
	Stroke	782 (9.39)	1.19 (1.09–1.31) ***
	Myocardial Infarction	464 (5.32)	1.38 (1.22–1.55) ****
	Heart Failure	832 (10.35)	1.69 (1.55–1.85) ****
	Death	498 (4.88)	1.82 (1.60–2.06) ****
BIDMC	Atrial Fibrillation	409 (4.75)	2.72 (2.31–3.21) ****
	Stroke	180 (1.92)	1.40 (1.12–1.74) ***
	Myocardial Infarction	130 (1.39)	1.10 (0.86–1.40)
	Heart Failure	520 (6.05)	2.33 (2.05–2.65) ****
	Death	378 (4.00)	1.65 (1.41–1.93) ****

Table 5. Hazard ratios with 95% confidence intervals for the EUH and BIDMC test sets, reflecting risk stratification based on the continuous neural network output (NN-Output). HRs correspond to the effect of a one–standard-deviation increase in NN-Output. Significance levels: *p < 0.05, **p < 0.01, ***p < 0.001, ****p < 0.0001.

Figure 3 shows the calibration between predicted risk scores from the Cox regression model based on NN-output and the observed 6-year risk. The ECI ranged from 0.02 to 0.12, with the poorest calibration observed for myocardial infarction in the EUH (0.08) and BIDMC (0.12) cohorts.

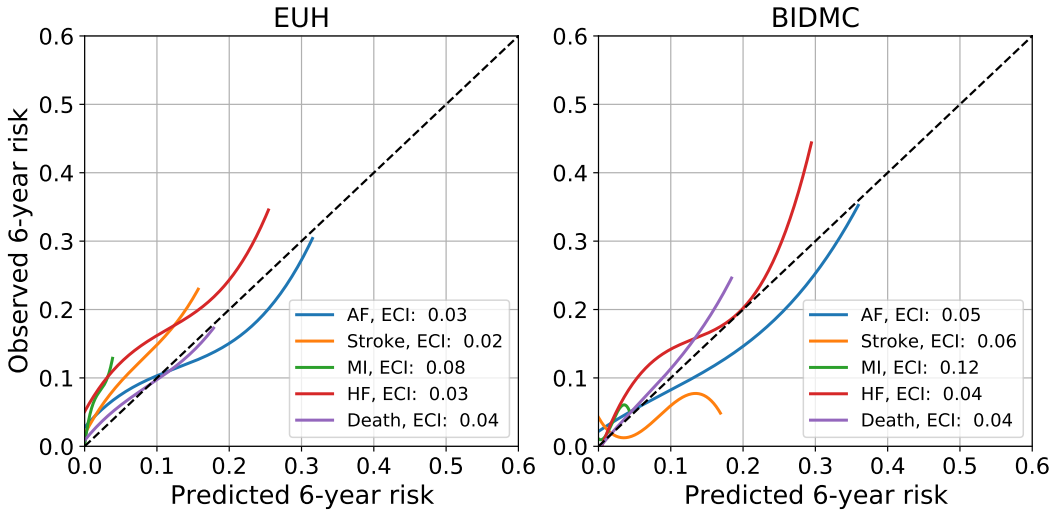


Figure 3. Calibration curves showing the relationship between predicted (based on NN-output) and observed 6-year risk for each cardiovascular outcome. Curves were fitted using a univariate spline with a smoothing factor of 1. The estimated calibration index (ECI, shown in the legend) represents the root mean squared difference between the mean predicted probabilities and the spline-fitted curve.

In the decision curve analysis (Figure 4), the covariates + NN-output model consistently demonstrated higher net benefit across a range of threshold probabilities, with substantial improvements over the covariates-only model for AF, HF, and death. For these outcomes, where incorporating the NN-output meaningfully enhances screening utility, we selected the classification threshold that yielded the highest net benefit for both EUH and BIDMC. At 6-year follow-up threshold of 0.07 resulted in sensitivities of 0.73 and 0.79 and specificities of 0.62 and 0.63 for AF in the EUH and BIDMC cohorts, respectively. For HF, the threshold 0.09 produced sensitivities of 0.71 and 0.79 and specificities of 0.63 and 0.67 for EUH and BIDMC. For death, with thresholdof 0.06, sensitivities were 0.71 and 0.74, with specificities of 0.62 and 0.70 for EUH and BIDMC, respectively. In contrast, for stroke, incorporating the NN output provided only a marginal benefit over the covariates-only model, while for myocardial infarction, it offered no additional net benefit compared to a screen-all strategy.

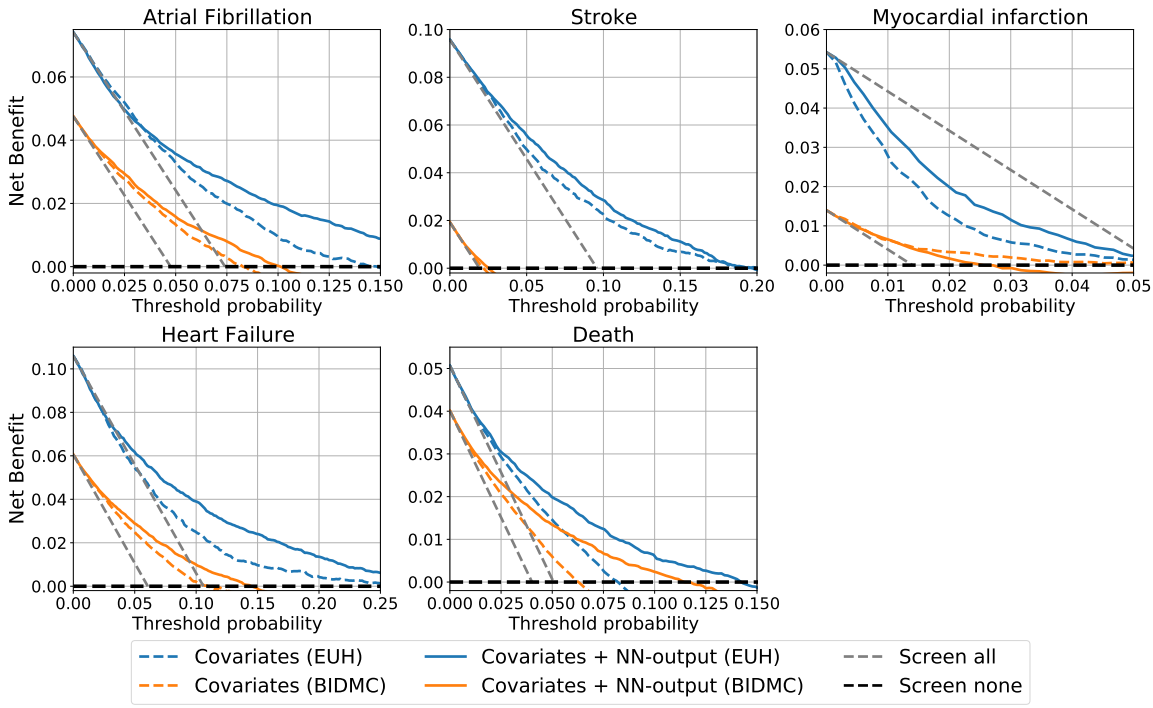


Figure 4. Decision curve analysis showing net benefit ($NB = \frac{TP}{N} - \frac{FP}{N}\frac{t}{1-t}$) plotted against the threshold probability, reflecting the trade-off between true positives and false positives at a 6-year time horizon. Curves correspond to models based on clinical covariates alone (dashed lines) and covariates augmented with the NN-output (solid lines), evaluated in the EUH (blue) and BIDMC (orange) cohorts. The "screen all" and "screen none" lines indicate strategies of screening all or no patients. Models with higher net benefit across a range of thresholds provide greater potential clinical usefulness. For myocardial infarction, inclusion of the NN-output did not improve net benefit compared with the "screen all" strategy across the evaluated threshold probabilities.

We also investigated what drives the model's predictions by examining how attention was distributed across 15-second ECG segments and across sleep stages in the test set. For each subject, we averaged the attention weights within each sleep stage and then compared these attention weights across the different cardiovascular outcomes. The resulting distributions are shown in Figure 5. Across all outcomes except MI, the model consistently assigned the highest attention to segments occurring during N1 sleep. Physiologically, N1 represents a transitional stage characterized by shifts in autonomic balance, reduced parasympathetic tone, and increased variability in cardiac electrophysiology. These rapid autonomic fluctuations may amplify subtle abnormalities in heart-rate dynamics or conduction patterns. The model's emphasis on N1 therefore suggests that early-stage sleep may contain particularly informative ECG signatures relevant to cardiovascular risk. It is important to note, however, that N1 is a light and brief stage that can sometimes be misclassified as wakefulness by expert scorers.

Finally, we examined the Spearman correlation coefficients between the NN-output and the covariates included in the Cox regression (Figure 6). Across all outcomes, the strongest correlations were observed with age and hypertension. Among the indices derived from expert sleep scoring, PLMI and time spent in N1 sleep showed the highest correlations with the NN-output. In contrast, time spent in N2, N3, and REM sleep was negatively correlated with the NN-output. These findings are consistent with the attention-weight analysis, which showed that the model placed greatest emphasis on N1 segments. With advancing

age, individuals typically experience reduced amounts of N2, N3, and REM sleep, accompanied by an increase in light sleep (N1). The positive correlation between N1 time and the NN-output therefore reinforces the relevance of the model's focus on this stage. Notably, sleep efficiency showed the strongest negative correlation with the NN output, suggesting that higher model-predicted cardiovascular risk is associated with poorer sleep quality. Specifically, individuals who later developed CVD tended to spend a smaller proportion of their time in bed asleep, resulting in lower sleep efficiency.

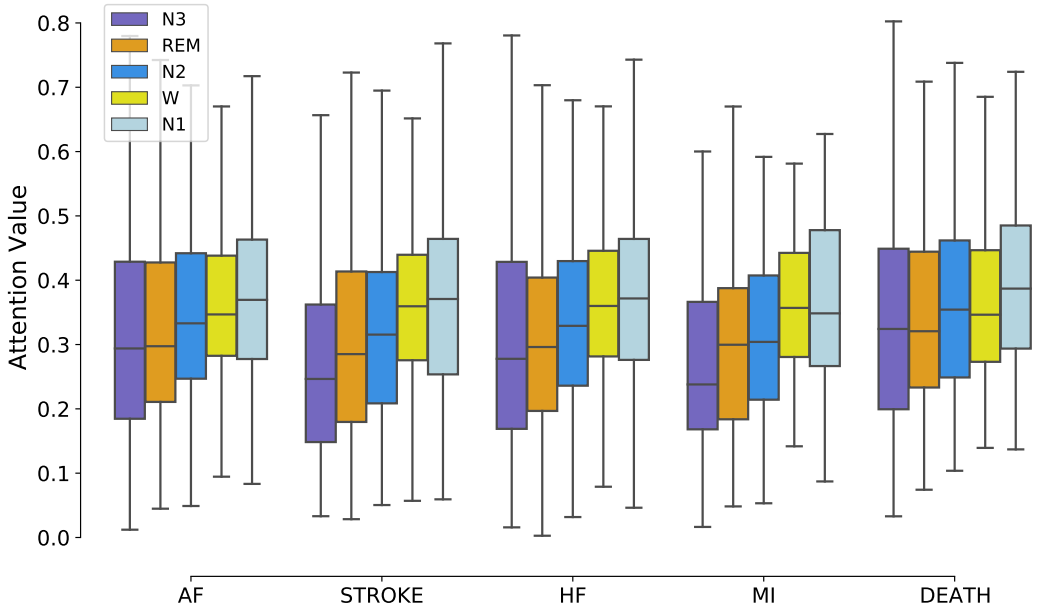


Figure 5. Attention weight distributions by sleep stage for the five predicted cardiovascular outcomes.

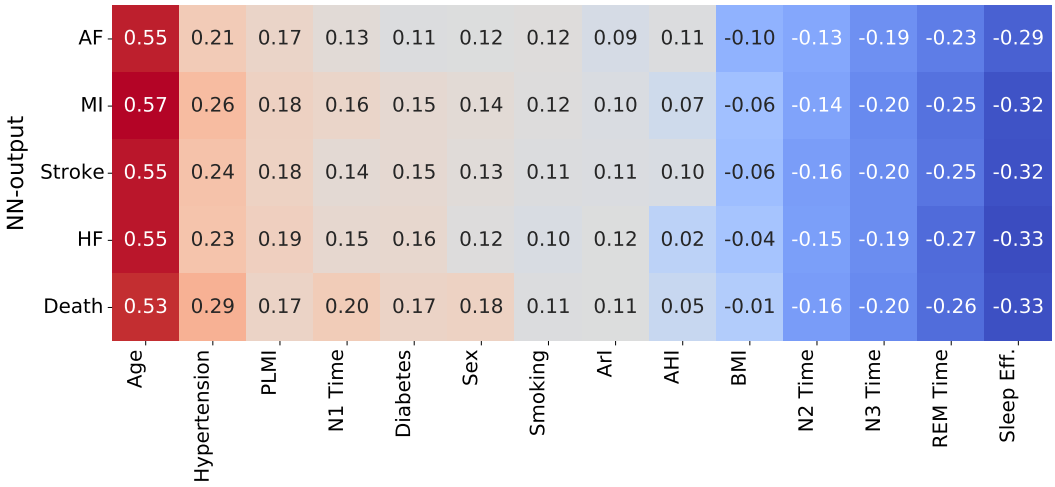


Figure 6. Spearman correlations of NN-output with Cox regression covariates across cardiovascular outcomes.

Sensitivity analysis results

Table 6 reports hazard ratios for subjects with at least two ICD codes corresponding to incident CVD. For AF and HF, hazard ratios increased compared with the primary analysis in both cohorts, indicating that the associations strengthen when using

more stringent outcome definitions. MI and stroke remained significant in the EUH cohort but not in BIDMC, likely due to the very low event rates in BIDMC (0.93 % for stroke and 0.60 % for MI).

Cohort	Outcome	Positive outcome (%)	Hazard ratios (95% CI)
EUH	Atrial Fibrillation	529 (6.30)	2.07 (1.83–2.32) ****
	Stroke	590 (7.27)	1.22 (1.09–1.36) ****
	Myocardial Infarction	344 (4.01)	1.40 (1.22–1.61) ****
	Heart Failure	698 (8.87)	1.78 (1.61–1.96) ****
BIDMC	Atrial Fibrillation	274 (3.23)	2.90 (2.37–3.53) ****
	Stroke	86 (0.93)	1.26 (0.93–1.72)
	Myocardial Infarction	55 (0.60)	1.21 (0.83–1.74)
	Heart Failure	355 (4.21)	2.61 (2.23–3.07) ****

Table 6. Hazard ratios with 95% confidence intervals for the EUH and BIDMC cohorts, reflecting risk stratification based on the continuous NN-output and including only patients with at least two ICD-9/10 codes corresponding to each cardiovascular outcome.

For the AF outcome, we performed an additional analysis excluding patients diagnosed with AF within 6 and 12 months after the PSG, as these individuals might have had paroxysmal AF that was not detected by our correction pipeline. Table 7 presents the corresponding hazard ratios, which decreased only modestly compared with the primary analysis.

	Positive outcome EUH (%)	Hazard ratios EUH (95% CI)	Positive outcome BIDMC (%)	Hazard ratios BIDMC (95% CI)
Atrial Fibrillation >6 months	490 (5.87)	1.97 (1.73-2.24) ****	296 (3.48)	2.35 (1.95 - 2.82) ****
Atrial Fibrillation >12 months	427 (5.15)	1.92 (1.70-2.16) ****	240 (2.84)	2.26 (1.85–2.76) ****

Table 7. Hazard ratios with 95% confidence intervals for atrial fibrillation based on NN-output, after excluding patients diagnosed with AF within 6 and 12 months following the PSG.

To evaluate the predictive value of the NN-output in a generally healthier population, specifically, individuals without any prior history of atrial fibrillation, stroke, myocardial infarction, or heart failure, we conducted a separate Cox regression analysis. The resulting hazard ratios for NN-output are summarized in Table 8.

Site	Outcome	Positive outcome (%)	Hazard ratios (95% CI)
EUH	Atrial Fibrillation	296 (4.65)	2.06 (1.77–2.41) ****
	Stroke	442 (6.96)	1.07 (0.95–1.20)
	Myocardial Infarction	199 (3.11)	1.21 (1.03–1.42) *
	Heart Failure	520 (8.18)	1.61 (1.45–1.80) ****
	Death	141 (2.22)	1.31 (1.06–1.61) **
BIDMC	Atrial Fibrillation	284 (3.70)	2.68 (2.21–3.34) ****
	Stroke	113 (1.47)	1.23 (0.94–1.60)
	Myocardial Infarction	73 (1.00)	1.12 (0.82–1.52)
	Heart Failure	376 (4.89)	2.19 (1.89–2.54) ****
	Death	146 (1.42)	1.24 (1.02–1.54) *

Table 8. Hazard ratios with 95% confidence intervals from a Cox proportional hazards model evaluating the predictive value of NN-Output for individuals without a history of heart disease (atrial fibrillation, stroke, myocardial infarction, or heart failure).

Discussion

In this study, we evaluated a neural network that integrates ECG signals with expert-annotated sleep stages to predict incident cardiovascular outcomes in patients undergoing overnight sleep studies. Model performance was assessed on two independent test sets including 9,810 patients from Emory University Hospital and 12,576 patients from Beth Israel Deaconess Medical Center. Model performance was evaluated using hazard ratios from Cox regression, calibration curves, and net benefit decision analysis.

Our analysis demonstrated that the neural network's predictions significantly stratified long-term cardiovascular risk across multiple endpoints as shown in Kaplan-Meier curves in Figure 2. The deep learning model also retained strong predictive value after adjustment for conventional cardiovascular risk factors (age, sex, BMI, diabetes, hypertension, and smoking) as well as sleep-related characteristics (AHI, ArI, PLMI, time spent in N1–N3 and REM sleep, and sleep efficiency) using Cox regression. A one-standard deviation increase in the NN-output was associated with substantially elevated hazard ratios: 2.03 for atrial fibrillation in EUH and 2.72 in BIDMC; 1.19 and 1.40 for stroke; 1.38 and 1.20 for myocardial infarction; 1.69 and 2.33 for heart failure; and 1.82 and 1.65 for all-cause mortality, respectively.

The NN-output derived predicted risk exhibited a robust agreement with observed risk for atrial fibrillation, stroke, heart failure, and all-cause mortality, as illustrated in calibration curves in Figure 3. In contrast, myocardial infarction showed higher estimated calibration indices, indicating poorer calibration. Decision curve analysis (Figure 4) demonstrated that incorporating NN-output into the Cox proportional hazards models yielded increased net benefit for atrial fibrillation, heart failure, and mortality. For stroke and myocardial infarction, inclusion of NN-output did not bring additional clinical benefit.

What we consistently see in hazard ratios, calibration plots, and decision curve analysis is that incorporating NN-output

into the prediction adds predictive value for AF, HF, and death, but not for MI and stroke. This might be explained by the lower prevalence of events for these CVDs, but also by the different nature of these conditions. While MI and stroke are cardiovascular diseases, they can occur due to vascular pathology anywhere in the circulatory system (coronary arteries for MI, cerebral vessels for stroke), making it more challenging to detect ECG-related features years before the actual incidence compared to primary cardiac conditions like AF and HF.

We also performed sensitivity analyses to increase the robustness of our outcome definitions. Specifically, we restricted the cohort to cardiovascular events supported by at least two corresponding ICD-9/10 codes to reduce potential EHR misclassification. After strengthening the diagnostic labels in this way, the hazard ratios for the NN-output, particularly for AF and HF, increased further, indicating strong prediction performance. When limiting the analysis to individuals without a prior history of cardiovascular disease (Table 8) neural network predictions remained significantly associated with an increased risk of AF, HF and death over the subsequent decade in EUH and over a 7-year follow-up in BIDMC. This suggests that the model can effectively predict future incidence even in populations with generally lower baseline risk.

Our study has several limitations. First, we predict cardiovascular outcomes solely based on ICD-9/10 codes, without direct clinical confirmation. We lack paired ECG recordings, one from the PSG sleep study and another from the time of a cardiovascular diagnosis, which would allow for a more comprehensive longitudinal analysis. Additionally, no biomarker data (e.g., N-terminal brain natriuretic peptide for heart failure or troponin I & T for myocardial infarction) are available, meaning our predictions rely entirely on the accuracy of ICD coding. Nevertheless, with a cohort of 38,195 patients, the noise in the ICD coding may be small compared to the cohort size, and sufficient to support our findings.

Another challenge is the potential misclassification of preexisting conditions, particularly AF. For instance, patients with undiagnosed paroxysmal AF may exhibit the arrhythmia during the PSG recording. In such cases, the model is not predicting future AF from subtle precursors but rather detecting existing, unrecognized AF during sleep. To address this, we included only patients whose first documented AF diagnosis occurred at least one month after the PSG acquisition, helping ensure that the model is truly predicting future onset rather than identifying AF already present in the signal. As an additional safeguard, we applied two automated AF classification algorithms to the ECG recordings and referred segments flagged as AF to a cardiologist for manual review. Due to resource limitations, only five 30-second ECG segments per full-night recording were examined, and only recordings associated with future AF diagnoses via ICD-9/10 codes were included. This approach enabled us to identify patients with previously undiagnosed AF at the time of PSG and exclude them from the prediction analysis, preventing overestimation of our model's predictive ability by avoiding inclusion of patients already experiencing AF. Given the large scale of our dataset—over 300,000 hours of PSG recordings—a full manual review was not feasible. However, combining ICD code-based selection with targeted review supports the validity of our prediction of future AF incidence. Additionally, sensitivity analysis excluding AF diagnoses within 6 and 12 months after PSG confirmed that higher NN-output predictions were associated with increased risk of atrial fibrillation, as reflected by the hazard ratios in Table 7.

Identifying individuals at highest risk for cardiovascular events is essential due to the substantial clinical and economic

burden of these conditions, as well as the availability of effective evidence-based interventions to mitigate risk[18,19]. Predicting cardiovascular outcomes from PSG using deep learning presents a non-invasive, scalable approach to assess long-term cardiac risk during routine sleep studies. Traditionally, cardiovascular risk assessment relies initially on an analysis of 12-lead ECGs, as demonstrated in heart failure and atrial fibrillation prediction models[20–22]. However, analyzing serendipitously collected nocturnal ECG signals using deep learning across an extended period of sleep offers a unique opportunity to detect subtle cardiac abnormalities that may not be apparent during wakefulness. Importantly, this approach also enables exploration of sleep stage–specific effects, as cardiac physiology varies significantly between REM and NREM sleep. This pattern is evident in the attention weights (Figure 5), which place greater emphasis on the N1 sleep stage, and in the Spearman correlation coefficients (Figure 6), where the NN-output is positively correlated with N1, but strongly negatively correlated with REM, N2, N3 sleep stages and sleep efficiency. Since many cardiovascular diseases are closely related to sleep disorders, integrating cardiovascular risk prediction into PSG analysis could improve early detection, enable personalized risk stratification, and potentially support more proactive and cost-effective patient care.

In contrast to most previous studies that rely solely on traditional cardiovascular risk factors, such as age, sex, diabetes, hypertension, smoking, and BMI[20–22], our approach integrates multiple complementary sources of information. First, we include the NN-output derived from overnight ECG recordings and sleep-stage information, allowing it to learn how ECG features vary across different stages of sleep. Second, in the Cox regression, we go beyond traditional covariates by also including sleep architecture metrics (e.g., time spent in N1, N2, N3, REM) and sleep-breathing indices such AHI, ArI, and PLMI. By integrating these measures, we can assess the influence of both neural network-derived cardiac features and sleep characteristics on cardiovascular outcomes. This approach emphasizes the critical role of sleep in cardiovascular risk, demonstrating that both the quality and architecture of sleep contribute meaningfully to disease prediction, in addition to traditional risk factors.

It is important to note that our study population does not represent the general population; however, this was not the intended scope of our work. Our analysis specifically focuses on PSG recordings from hospital-based sleep studies, targeting individuals with sleep disorders who are often at elevated risk for cardiovascular disease. In the sleep studies, the ECG is typically not the primary focus, and our objective was to assess its predictive value for cardiovascular outcomes within this specialized population rather than in a general cohort.

We recognize that false-positive predictions could have important clinical and psychological consequences, including unnecessary downstream testing, increased healthcare costs, and patient anxiety. However, prior real-world deployments of AI-enabled ECG have demonstrated that such tools can be used effectively as screening instruments within structured clinical workflows, rather than as standalone diagnostic tests, thereby limiting unnecessary interventions and maintaining cost-effectiveness[23,24]. In a potential clinical deployment scenario, NN-output predictions combined with traditional risk factors could be used as a screening tool, with positive predictions prompting confirmatory evaluation using standard clinical assessments, such as follow-up ECGs, Holter monitoring, or cardiologist review to help limit unnecessary interventions.

Conclusion

In this study, we presented a deep learning approach for predicting cardiovascular outcomes, including atrial fibrillation, stroke, myocardial infarction, heart failure, and death. By leveraging a large dataset of ECG recordings from sleep studies, our method demonstrates the potential to identify individuals at high risk for cardiovascular disease years before clinical symptoms arise. Our findings indicate that predicting cardiovascular outcomes from the ECG channel in polysomnography is a promising approach, with deep learning capable of extracting valuable subtle information from ECGs, which are often not the primary focus in sleep studies due to the emphasis on neurological and other physiological signals. This study demonstrates the potential of predicting atrial fibrillation, heart failure, and mortality from overnight PSGs, whereas predictions for myocardial infarction and stroke require further optimization, as the deep learning model did not substantially improve risk prediction upon established risk factors. With continued development, our method has the potential to serve as a non-invasive screening tool that could be integrated into existing diagnostic protocols for patients undergoing sleep studies, particularly for the assessment of AF, heart failure, and mortality risk.

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Disclosure of Interest

Dr. Westover is a co-founder, scientific advisor, consultant to, and has personal equity interest in Beacon Biosignals.

Data Availability Statement

The data used in this study are part of the Human Sleep Project (https://bdsp.io/content/hsp/2.0/) hosted on the Brain Data Science Platform and are currently available under restricted access. By requesting access to the data, you agree not to download, copy, repost or to publish or otherwise share any work that uses the data, in full or in part, without written consent from the BDSP leadership team. This condition will be loosened at a later date.

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